

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/CSE/BS-M 401/2025-26
2026
MATHEMATICS - III

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5 x 2 = 10

1. Does there exist a simple graph with 5 vertices having degrees 2, 2, 4, 4, 4? 2
2. Find the limit, as $n \rightarrow \infty$, of the sequence $\{a_n\}$ where $a_n = \frac{x^n}{n!}$. 2
3. Evaluate $\int_0^1 dy \int_y^1 e^{x^2} dx$. 2
4. Find Integrating factor of the differential equation
 $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)xdy = 0$ 2
5. If $u = \log_e(x^3 + y^3 + z^3 - 3xyz)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$ 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *five* questions

12 x 5 = 60

6. i) Test the convergence of the series

4

$$\frac{x}{1} + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1.3}{2.4} \cdot \frac{x^5}{5} + \frac{1.3.5}{2.4.6} \cdot \frac{x^7}{7} + \dots \dots \dots (x > 0).$$

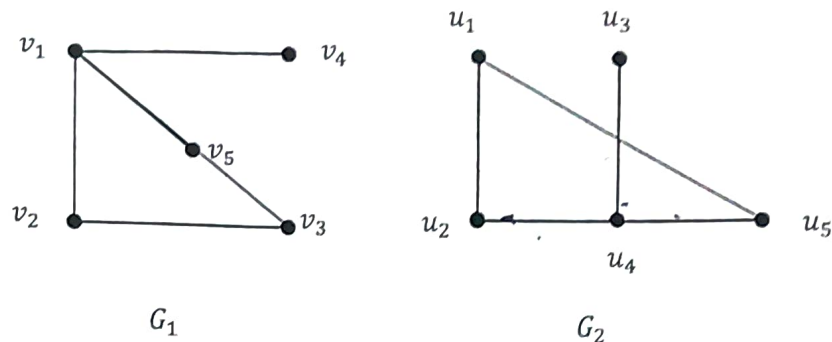
- ii) Prove that the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \dots \dots$ is absolutely convergent 4
when $|x| < 1$ and conditionally convergent when $x = 1$.

- iii) If $a_n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots \dots \dots + \frac{1}{n(n+1)}$, show that $\{a_n\}$ is a bounded 4
monotonic increasing sequence.

7. i) Show that the maximum number of edges in a simple graph with n vertices and m components is $\frac{1}{2}(n-m)(n-m+1)$. 4

- ii) Show that a tree with n vertices has size $(n-1)$. 4

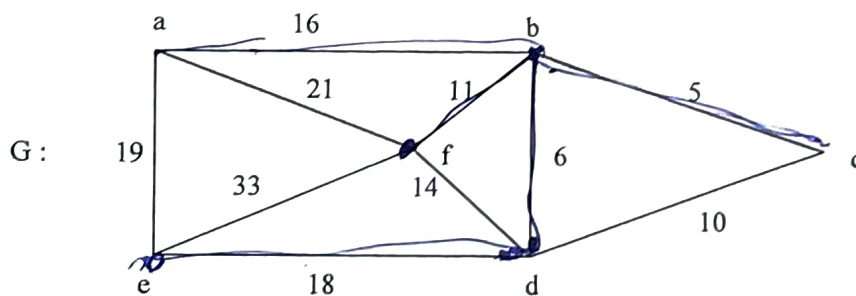
- iii) Find the adjacency matrices of the graphs G_1 and G_2 : 1+1+2



Examine whether the graphs are isomorphic or not .

8. i) Show that any tree with $n(\geq 2)$ vertices has at least two pendant vertices. 4
 ii) Define a spanning tree T and the co-spanning tree T^* of a connected graph G . 2+2
 iii) Show that every connected graph has at least one spanning tree. 4

Use Prim's algorithm to find the minimal spanning tree in the graph G given below :



9. i) Solve: $y = 2px + 4xp^2$, where $p = \frac{dy}{dx}$. 4
 ii) Solve: $\frac{dy}{dx} + y = y^3(\cos x - \sin x)$ 4
 iii) Solve: $(x^2 + x)p^2 + (x^2 + x - 2xy - y)p + y^2 - xy = 0$, where $p = \frac{dy}{dx}$. 4
 10. i) Obtain the general solution and the singular solution of the equation $p = \cos(y - px)$, where $p = \frac{dy}{dx}$. 4
 ii) Solve: $(D^2 + 1)y = x \cos x$, where $D \equiv \frac{d}{dx}$. 4

- iii) Apply the method of variation of parameters to solve :

4

$$(D^2 + 4)y = \tan 2x, \text{ where } D \equiv \frac{d}{dx}.$$

11. i) Solve the equation

6

$$(3x + 2)^2 D^2 y + 3(3x + 2)Dy - 36y = 3x^2 + 4x + 1, \text{ where } D \equiv \frac{d}{dx}.$$

- ii) Verify the divergence theorem for the vector function

6

$$\vec{F} = 4xzi - y^2j + yzk \text{ taken over the cube bounded by}$$

$$x = 0, x = 1; y = 0, y = 1; z = 0, z = 1.$$

12. i) Give the function $f(x, y) = xy$ if $|x| \geq |y|$

4

$$= -xy \text{ if } |x| \leq |y|$$

Prove that $f_{xy}(0,0) \neq f_{yx}(0,0)$.

- ii) If $w = f(2x - 3y, 3y - 4z, 4z - 2x)$, Prove that

4

$$\frac{1}{2} \left(\frac{\partial w}{\partial x} \right) + \frac{1}{3} \left(\frac{\partial w}{\partial y} \right) + \frac{1}{4} \left(\frac{\partial w}{\partial z} \right) = 0.$$

- iii) Apply Stokes theorem to evaluate

4

$$\int_C (ydx + zdy + xdz) \text{ where } C \text{ is the curve of intersection of}$$

$$x^2 + y^2 + z^2 = a^2 \text{ and } x + z = a.$$

13. i) Find the extreme of the function

4

$$f(x, y, z) = x^2y^2 - 5y^2 - 8xy - 5x^2.$$

- ii) Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction of $2i - j - 2k$.

4

- iii) Evaluate $\iiint (x + y + z)^4 dx dy dz$ over the three dimensional region $x \geq 0, y \geq 0, z \geq 0$, and $x + y + z \leq 1$.

4

Object Oriented Programming

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

- | | |
|--|--------|
| 1. What do you mean by constructor and destructor? | 5x2=10 |
| 2. Justify the statement that a final class is a complete class. | [2] |
| 3. What do you mean by virtual function and pure virtual function? | [2] |
| 4. What is a copy constructor? | [2] |
| 5. Define Is-a relationship and Has-a relationship. | [2] |

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15 = 60

- | | |
|---|---------|
| 6. i) What do you mean by procedural-oriented programming and object-oriented programming paradigms? Explain different features of object-oriented programming. | [2+5] |
| ii) How are different static members defined in a class and accessed? Explain with suitable programming code. | [6] |
| iii) What are the purposes of the super keyword? | [2] |
| 7. i) Explain the different levels of access modifiers in terms of packages available in Object Oriented Programming with suitable programming code. | [8] |
| ii) What is an interface? Explain how to define an interface and implement it using a class with suitable programming code. | [1+4] |
| iii) Explain the purpose of 'this' keyword. | [2] |
| 8. i) Explain different forms of inheritance in Object Oriented Programming. | [6] |
| ii) What is constructor chaining? Explain the execution of constructors and destructors in hierarchical inheritance and multiple inheritance with suitable examples. | [1+5] |
| iii) Define friend function and friend class in Object Oriented Programming. | [3] |
| 9. i) What is a thread? Why is it called a lightweight process? Explain the life cycle of a thread object with a suitable block diagram. | [1+1+4] |
| ii) Explain the different ways to create a user-defined thread class with suitable programming code. | [6] |
| iii) Develop a simple application program to illustrate the use of multithreads. | [3] |
| 10. i) What is operator overloading? Overload the '+' and '*' operators using non-member functions to perform addition and multiplication of two complex numbers, respectively. | [1+6] |
| ii) What is the main drawback of multiple inheritance? How can it be avoided by using a virtual class? | [1+4] |
| iii) Write a program where the interface can be used to support multiple inheritance. | [1] |

FORMAL LANGUAGE AND AUTOMATA THEORY

Full Marks: 70

Times: 3 Hours

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GROUP-A [OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

1. Define DFA. 5x2=10
2. Define Regular Grammar with example. 2
3. Describe two normal forms of Context Free Grammar. 2
4. Define four Identities of Regular expressions. 2
- Find the regular expressions over $\{a,b\}$ for (i) Set of all strings containing exactly 2a's (ii) The set of all strings containing aa as substring. 2

GROUP-B [LONG ANSWER TYPE QUESTIONS]

Answer any **four** questions

6. (a) Differentiate between DFA and NFA. Construct a DFA equivalent to the NFA given in Figure 1. 4x15 = 60
- (b) Design a DFA which accepts a set of all strings containing an even number of zeros and an odd number of ones. Show the transition sequence to accept the string 0010110. 4+6

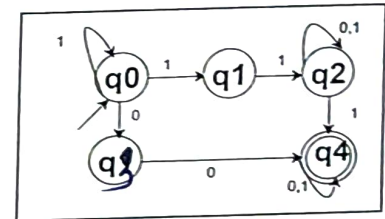


Figure 1.

7. (a) What do you mean by equivalent states? Construct the minimum state automata equivalent to the Transition Table given in Table 1 where F,G,H are the final state. Show the state transition table and transition diagram. 1+8
- (b) Find the language accepted by the Finite Automata in Figure 2. If the final states and non-final states are interchanged, then which language will be accepted by the new DFA? 2
- (c) Construct a transition diagram that recognizes the set $R = (1(00)^*1 + 01^*0)^*$. 4

Present State	Next State	
	I/P = a	I/P = b
A	B	F
B	A	F
C	G	A
D	H	B
E	A	G
F	H	C
G	A	D
H	A	C

Table 1

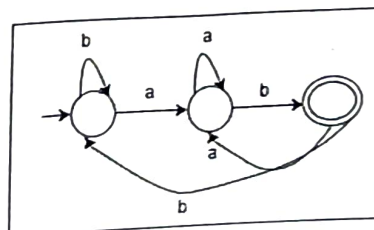
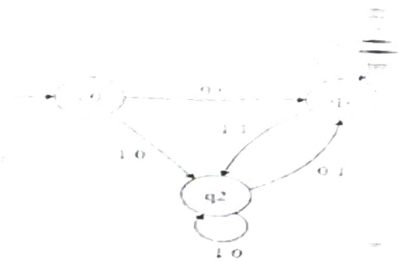


Figure 2.

8. (a) Define Mealy Machine and Moore. Convert the following Mealy machine to Moore Machine.
 (b) Define the different types of grammars according to the Chomsky classification and write the languages accepted by each category of grammars.
 (c) If $S \rightarrow aS \mid a$, Find $L(G)$



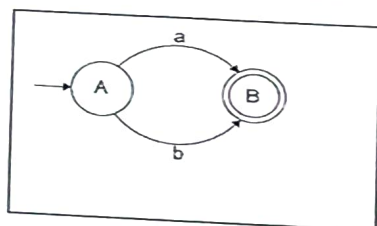
3+5

6

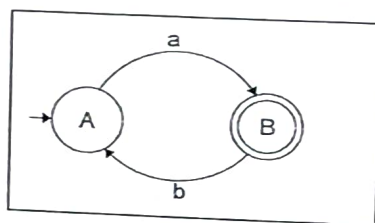
1

9. (a) Define pumping Lemma for regular language. Using the pumping lemma proof that $a^i b^i$, $i \geq 1$ is not regular. 3+5
 (b) Reduce the following grammar to Chomsky Normal Form: $S \rightarrow ASA \mid bA$, $A \rightarrow B \mid S$, $B \rightarrow c$ 7

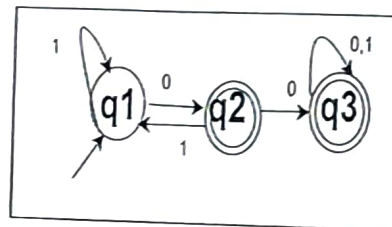
10. (a) Find a grammar generating $L = \{ a^n b^n c^i \mid n \geq 1, i \geq 0 \}$ 3
 (b) Find Regular expressions for the following FA in Figure (i) and Figure (ii) 2
 (c) Solve the automata using Arden's Theorem in Figure (iii). 6
 (d) Consider the grammar G given by $S \rightarrow 0SA_12$, $S \rightarrow 012$, $2A_1 \rightarrow A_12$, $1A_1 \rightarrow 11$. Find the Language $L(G)$. Show the derivations to check whether the string 001122 is accepted by $L(G)$. 2+2



(i)



(ii)



(iii)

11. (a) What is a parse tree? Construct a parse tree for the string "bbabba" accepted by the Grammar: $S \rightarrow bSb \mid a \mid b$ 1+2
 (b) Let $G = (\{A_0, A_1\}, \{a, b\}, P, A_0)$, where P consists of $A_0 \rightarrow aA_1$, $A_1 \rightarrow bA_1$, $A_1 \rightarrow a$, $A_1 \rightarrow bA_0$. Construct a transition system accepting $L(G)$. 3
 (c) Consider the NFA with ϵ (null) transitions into DFA where $M = (\{q_1, q_2, q_3, q_4\}, \{a, b\}, \delta, q_1, q_4, F)$ and $\delta(q_1, a) \rightarrow q_2$, $\delta(q_1, \epsilon) \rightarrow q_3$, $\delta(q_2, b) \rightarrow q_3$, $\delta(q_2, \epsilon) \rightarrow q_2$, $\delta(q_3, a) \rightarrow q_3$, $\delta(q_3, a) \rightarrow q_4$, $\delta(q_4, b) \rightarrow q_1$, $\delta(q_4, b) \rightarrow q_4$ 7
 (d) Give definition of Kleen's closure. 2

1+2

3

7

2

3

2+1

5

1+3

12. (a) Differentiate between the closure properties of Regular Grammar and Context Free Grammar.
 (b) Define push down automata. How can we define the acceptance of a string by a push down automata. Construct a pda accepting $L = \{wcw^T \mid w \in \{a, b\}^*\}$
 (c) What is meant by ambiguity in CFG. Show that the following grammar is ambiguous:
 $S \rightarrow S + S$, $S \rightarrow S * S$, $S \rightarrow a$, $S \rightarrow b$

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
COE/B.TECH./CSE/PCC-CS401/2025-26
2026
COMPUTER ARCHITECTURE

24101104009

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer all questions

5x2=10

1. "Synchronization of all PEs in an SIMD computer is done by hardware rather than by software" - justify the statement.
2. Define Belady's Anomaly with an example.
3. Define the vector gather and vector scatter instructions with examples.
4. "A superpipelined processor performs less effectively than a superscalar processor of small degree" - Explain it.
5. What are the differences between UMA and NUMA?

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15=60

6. i) Discuss the Flynn's Taxonomy of computer architecture. Why the MIMD processor is more reliable than the SIMD processor? 4+1
- ii) Explain the inclusion property and memory coherence requirements in a multilevel memory hierarchy. Distinguish between write-through and write-back policies in maintaining the coherence in adjacent levels. 2+2
- iii) Consider a two-level memory hierarchy, M1 and M2, with access times t_1 and t_2 , costs per byte c_1 and c_2 , and capacities s_1 and s_2 , respectively. The cache hit ratio $h_1 = 0.95$ at the first level.
 - (a) Derive a formula showing the effective access time t_{eff} of this memory system. (b) Derive a formula showing the total cost of this memory system. 1+1
 - (c) Suppose $t_1 = 20$ ns, t_2 is unknown, $s_1 = 512$ Kbytes, s_2 is unknown, $c_1 = \$0.01/\text{byte}$, and $c_2 = \$0.0005/\text{byte}$. The total cost of the cache and main memory is upper-bounded by \$15,000.
 - (i) How large a capacity of M2 ($s_2 = ?$) can you acquire without exceeding the budget limit? 2
 - (ii) How fast a main memory ($t_2 = ?$) do you need to achieve an effective access time of $t_{eff} = 40$ ns in the entire memory system under the above hit ratio assumptions? 2
7. i) Define vector loops and vector chain with example. 3
- ii) The following sequence of compound vector function (CVF) is to be executed on a Cray X-MP vector processor with two vector-load pipes and one vector-store pipe which can be used simultaneously with the remaining functional pipelines.

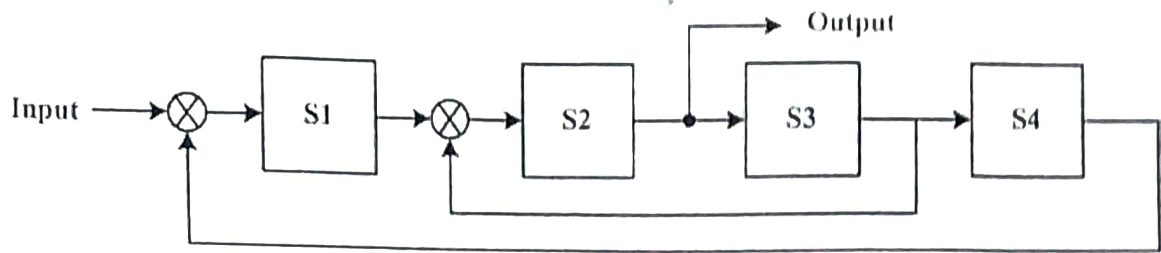
$$A(I) = B(I) + s \times C(I)$$

$$D(I) = s \times B(I) \times C(I)$$

$$E(I) = C(I) \times (C(I) - B(I))$$
 where $B(I)$ and $C(I)$ are each 64-element vectors originally stored in memory. The resulting vectors $A(I)$, $D(I)$, and $E(I)$ must be stored back into memory after the computation.
 - (a) Write 11 vector instructions in proper order to execute the above CVFs on the above-mentioned processor. 3
 - (b) Show the pipeline-chaining diagram for executing the above CVF on a Cray X-MP vector processor. 3
 - (c) Show a space-time diagram, for achieving maximally chained vector operations for executing the above CVFs in minimum time. 4
 - (d) Show the potential speedup of the above vector chaining operations over the chaining operations on the Cray 1, which has only one memory-access pipe. 3
- iii) What are the main differences between register to register and memory to memory architecture of vector processor? 2

- 8
- Differentiate data flow computer from control flow computer?
 - List two potential problems with data flow computer implementation.
 - Use Bernstein's conditions to detect the parallelism embedded in the following code:
 $1: C = D * E; \quad 2: M = G + C; \quad 3: A = B + C; \quad 4: C = L + M; \quad 5: F = G / E$
 Also draw a data flow graph for the above code.
 - A nonpipelined processor X has a clock rate of 25 MHz and an average CPI (cycles per instruction) of 4. Processor Y, an improved successor of X, is designed with a five-stage linear instruction pipeline. However, due to latch delay and clock skew effects, the clock rate of Y is only 20 MHz.
 - If a program containing 100 instructions is executed on both processors, what is the speedup of processor Y compared with that of processor X?
 - Calculate the MIPS rate of each processor during the execution of this particular program.

- 9
- Consider the following pipelined processor with **four stages**. This pipeline has a total evaluation time of **six clock cycles with $t = 25$ ns**. All successor stages must be used after each clock cycle.



- Specify the reservation table for this pipeline with six columns and four rows.
 - List the set of forbidden latencies and state the initial collision vector.
 - Draw the state diagram which shows all possible latency cycles.
 - List all simple cycles and greedy cycles from the state diagram.
 - What is the value of the minimal average latency (MAL)?
 - What is the maximal throughput of this pipeline?
- Identify all the RAW, WAR and WAW hazards in the instruction sequence:
 $DIV R1, R2, R3; SUB R4, R1, R5; ADD R2, R6, R7; MULT R8, R4, R2; OR R3, R11, R1$
- 10
- Explain the basic concepts of paging and segmentation in managing the physical and virtual memories in a hierarchy with diagrams.
 - A two-level memory system has eight virtual pages on a disk to be mapped into four page frames (PFs) in the main memory. A certain program generated the following page trace:
 $1, 0, 2, 2, 1, 7, 6, 7, 0, 1, 2, 0, 3, 0, 4, 5, 1, 5, 2, 4, 5, 6, 7, 6, 7, 2, 4, 2, 7, 3, 3, 2, 3$
 - Show the successive virtual pages residing in the four page frames with respect to the above page trace using the LRU replacement policy. Compute the hit ratio in the main memory. Assume the PFs are initially empty.
 - Repeat part (a) for the Optimal page replacement policy. Compute the hit ratio in the main memory.
 - Find the size of logical and physical address for system with a logical address space of 16 pages of 1024 words each, mapped onto a physical memory of 64 frames.
- 11.
- Draw the memory hierarchy and explain it.
 - Write the formula of the following: a) Speed up factor (S_k) of pipelining b) Efficiency of pipeline processor, c) throughput of pipeline processor, d) effective access time of memory hierarchy, e) hit ratio and f) inclusion in memory hierarchy.
 - Show the data flow execution of $X = (a + b) / (a - b)$ considering $a=10, b=5$.
 - Explain different types of internal data forwarding to improve the throughput of the pipeline processor.
 - Apply internal data forwarding on the following program:
 $R1 \leftarrow M[A]; R2 \leftarrow M[A]; R1 \leftarrow R1 * R2; R3 \leftarrow M[B]; R1 \leftarrow R1 + R3; M[A] \leftarrow R1;$

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

1. What is the difference between explicit and implicit constraints?
2. Which asymptotic notation is used widely in algorithm analysis, and why?
3. What are heuristics? Do they always guarantee solutions?
4. "Greedy method always gives an optimal solution" - Justify your answer.
5. Do solutions exist for one, two, and three-queen problems? - Justify your answer.

5x2=10

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

6. i) Given the weight vector (15,25,35,45,55) and the profit vector (10,20,30,40,50) and a knapsack of capacity 100, find the optimal solution for the knapsack problem of five objects
ii) Explain any single-source shortest path algorithm. Also, calculate its time complexity.
iii) How will you solve the 0/1 knapsack problem using a dynamic programming paradigm?
7. i) Discuss Strassen's matrix multiplication algorithm to evaluate the product of nxn matrices. Find the resulting recurrence relation for the same and calculate its time complexity.
ii) Explain the Travelling Salesperson problem with an example and solve it with dynamic programming concept.
8. i) What is Turing's halting problem? Prove that the halting problem is undecidable.
ii) How does the Floyd-Warshall algorithm work? - Explain. Calculate its time complexity.
iii) What do you mean by the rate of growth of functions? What are coding standards? Give its utility for writing the code.
9. i) Explain the algorithm development life cycle (ADLC) with examples in each phase.
ii) Write down the recursive approach of any backtracking algorithm.
iii) Why is the average case time-complexity of the quicksort algorithm is $O(n \log n)$? Prove it.
10. i) Which strategy will you follow to solve the 15-puzzle problem? Also, draw the necessary steps required to solve the problem.
ii) If $f(n) = a_m n^m + a_{m-1} n^{m-1} + \dots + a_1 n + a_0$, where $a_m > 0$, then show that $f(n) = O(n^m)$.
iii) Differentiate between backtracking and branch and bound technique? Calculate the complexity of merge sort algorithm in all three cases.
11. i) Write down Prim's algorithm to find the minimal spanning tree of an undirected graph. Compare and contrast Prim's and Kruskal's algorithm.
ii) Find the minimum number of operations required for the following matrix chain multiplication using dynamic programming: $A(10 \times 20) * B(20 \times 15) * C(15 \times 30) * D(30 \times 5)$.
iii) Discuss the different types of complexity classes of problems.
12. i) What is the principle of optimality condition? How is it satisfied in the graph coloring problem?
ii) Trace the steps to solve the 4-Queens problem by the backtracking method. For each step, draw the 4x4 matrix showing the positions of queens in it. Show where you apply backtracking. Draw the portion of the state space tree for n = 4 queens using the backtracking algorithm.

4x15=60

5

4+1

5

5+5

5

2+3

4+1

3+1+1

5

5

5

5

5

2+3

3+2

7

3

2+3

END

2+2 2+4

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JGEC/B.TECH. /CSE/MC-401/2025-26
2026
ENVIRONMENTAL SCIENCE

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

1. Write down the main objectives of Environmental Science.
2. What is the doubling time of a population?
3. Define the term 'Eutrophication'.
4. What is a catalytic converter?
5. Name two gases responsible for acid rain.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

Answer any **four** questions

6. i) What is meant by energy resources? Discuss the various types of energy resources in detail. 2+5
ii) What are the threats associated with energy resources? 4
iii) How can we overcome the energy crisis? 4
7. i) State the term 'thermal pollution'. How does thermal pollution affect the environment, and how can it be controlled? 2+5+5
ii) What is meant by primary and secondary pollutants? 3
8. i) What is meant by the hardness of water? What are the different types of hardness? Discuss the processes used to remove hardness from water. 2+2+5
ii) Write down the importance of dissolved oxygen. Discuss the nitrogen cycle. 3+3
9. i) Describe the structure of the atmosphere with reference to temperature variation in its different layers. 5
ii) What is ozone layer depletion? Discuss its causes and harmful effects on living organisms and the environment. 2+2
iii) Explain the greenhouse effect. How does it contribute to global warming? 3+3
10. i) Define air pollutants. Differentiate between primary and secondary air pollutants using suitable examples. 1+4
ii) What is particulate matter (PM)? Discuss its sources and effects on human health. 1+4
iii) What is smog? Classify smog and explain the characteristics of each type. 3x5
11. Write short note on **any three** of the following:
(i) Food chain (ii) BOD and COD (iii) Floods and Landslides (iv) Exponential population growth model (v) PAN (Peroxyacetyl nitrate) (vi) Electrostatic precipitator